Shaurya Mittal

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Education

University of Waterloo 2025 – Present
Bachelor of Computer Science (Honours) Co-op Program *Waterloo, ON*

Skills

Languages: Python, TypeScript, JavaScript, Go, Java, C, SQL
AI & LLMs: Claude API, OpenAI API, RAG, LLM Evaluation (LLM-as-judge), Vector Databases (Qdrant), MCP
Automation & Agents: Glean Agents & Workflows, Zapier, n8n
Frameworks & Libraries: React, Next.js, Node.js, Express, Streamlit
Cloud & Data: AWS (S3, Athena, CloudWatch), Snowflake, PostgreSQL, MongoDB, Firestore, Firebase
Tools: Git, Docker, GitLab CI/CD, REST APIs, Stripe API, Chrome Extensions (Manifest V3)

Experience

AI Solutions Engineering Intern May 2026 – August 2026
Docebo Inc. *Toronto, ON*

- Built a full-stack agent feedback dashboard (React, Glean OAuth, AWS CloudWatch/S3/Athena) giving agent owners role-scoped, self-service access to user feedback, replacing manual data pulls by AI Operations.
- Built 20+ Glean agents and workflows for teams across the company, and ran pilot evaluations of vendor AI and automation platforms, delivering reports comparing their workflow-building and agent capabilities.
- Engineered a company-wide AI portal (Next.js, MCP server, GitLab-backed) serving 900+ employees as a single hub for AI guides, the full agent catalog, and self-service skill and agent registration.
- Diagnosed a silent pagination bug truncating Glean API data and shipped a reconciliation sync (capped at 30% deletions per run) for an 800+ agent catalog; audited it end-to-end, removing 670+ stale or duplicate agents (67%).

Projects

hookfuzz – Chaos Testing CLI for Payment Webhooks September 2026
Go, Node.js, Express, Stripe API | github.com/shaurmittal/hookfuzz

- Built a Go CLI that finds bugs in Stripe webhook handlers by delivering signed events twice, out of order, late, or after failures, then checking user-defined correctness rules against the app's final state.
- Implemented delta-debugging test shrinking with seeded runs, reducing a 44-event failing sequence to a minimal 2-event reproduction that replays exactly from a single seed.
- Validated against a sample Express shop with 3 planted bugs (duplicate fulfillment, dropped refund, stale subscription overwrite): caught all 3 with zero false positives over 200 seeds; misconfigurations fail loudly instead of reporting a false pass.

Countersign – AI Payment Fraud Defense for Small Businesses October 2026
TypeScript, Next.js, Convex, Claude API, Cartesia Voice | github.com/shaurmittal/countersign-calhacks

- Built a second-signer system at Cal Hacks 13.0 that verifies outbound payment requests before money moves, keeping LLMs off the authorization path: Claude parses requests and flags injection, while a deterministic, unit-tested TypeScript policy engine makes every pay/refuse decision with no model calls.
- Largest contributor on a team of 4 (82 of 152 commits), owning multi-channel intake across voice (Cartesia), email, WhatsApp, and web chat, plus the out-of-band callback agent that verifies vendors on the number already on file rather than the one in the request.
- Authenticated email on real SPF/DKIM/DMARC results via Postmark Inbound rather than the displayed sender, defeating forged vendor and owner emails; shipped the operator Next.js UI, backed by a 39-scenario CI suite (26/26 scams blocked, 13/13 legitimate bills handled) over 1,126 tests.

FastAPI RAG Q&A Engine September 2026
Python, Claude API, Qdrant, Streamlit, Docker | github.com/shaurmittal/rag-fastapi-eval

- Built a cited-answer RAG system over FastAPI's docs and source using Qdrant, local bge-small embeddings, a cross-encoder reranker, and Claude with native citations, deployed as a Streamlit demo.
- Ran a 4-configuration ablation (chunking × reranking) on 51 questions mined from 613 closed GitHub issues; paired bootstrap CIs showed reranking cost ∼15× latency for no significant gain.
- Built an LLM-as-judge pipeline scoring claim-level faithfulness and citation precision, validated against an independent rater ($\kappa = 0.89$); a pooled TREC-style label review then recovered 28 answers the first pass missed, raising hit@5 by ∼0.10 – a larger effect than any pipeline change.